

Role prompting

💡 Idea

- Role prompting is a technique where multiple "agents" are instances of the same underlying LLM, distinguished only by a system prompt that casts each in a named role — for example "You are a mathematician. You are good at math games, arithmetic calculation, and long-term planning."
- Each role-prompted agent is sampling from a different conditional distribution over outputs of one shared model. The "agents" share parameters; only their context differs.

❓ Why It Matters

- It is the dominant way to construct heterogeneous-looking agent teams in LLM multi-agent systems — cheap, no [fine-tuning](#) needed, easy to vary.
- The diversity benefit of a multi-agent setup depends entirely on whether the role-induced conditional distributions are different enough in content (not just in style) to be worth aggregating. If they collapse to the same answers, the "team" reduces to one model voting multiple times.

⊗ Not This

- Not the same as using different models (different architectures or different fine-tuned checkpoints). Role-prompted agents share all weights.
- Not the same as a chain-of-thought prompt — a role prompt sets a persona system message; CoT shapes the reasoning trace within one persona.

☰ Examples

- [Liu2024ADL](#)'s general-reasoning agents on MMLU: Mathematician, Programmer, Lawyer, Historian, Economist, Psychologist, Doctor — all instantiated from the same gpt-35-turbo-0301 backbone, distinguished only by their system prompt ([app. D, tab. 16, p. 27](#)).

- [Liu2024ADL](#)'s code-generation agents: Python Assistant, Algorithm Developer, Computer Scientist, Programmer — all role-prompted instances of the same backbone ([app. D, tab. 16, p. 28](#)). The same setup also includes Syntax Checker (a pure tool, no LLM) and Unit Tester (an [LLM agent](#) that writes unit tests); these sit alongside the role-prompted agents in the agent pool.
- [Chen2024AgentVerseFM](#)'s recruiter-generated roles: a "recruiter" LLM is prompted with the goal and emits role descriptions like "an electrical engineer specified in the field of xxx", "an economist who is good at xxx", which then become the role prompts for that round's agents ([fig. 7, p. 27](#)). The pool itself is generated per goal, not picked from a fixed list.
- [ChatDev](#)'s five roles — CEO, CTO, programmer, reviewer, tester — are all the same frozen ChatGPT-3.5, differentiated only by system messages. Here the pairing of an instructor prompt with a matched assistant prompt is the specialized form called [inception prompting](#).
- [ReCon](#)'s six [Avalon](#) players run on one shared backbone (ChatGPT or Claude-2), differentiated only by role-specific system prompts and hints that give Merlin, Percival, Loyal Servant, Morgana, and Assassin distinct knowledge and goals ([Wang2023AvalonsGO](#); [app. E.2, fig. 16, p. 22](#)).
- [Werewolf](#)'s seven players in [Xu2023ExploringLL](#) run on one shared gpt-3.5-turbo-0301 backbone, differentiated only by role prompts that give each role ([werewolf](#), villager, seer, guard, witch) its distinct abilities and objectives ([sec. 3.2, p. 3](#)).
- [MetaGPT](#)'s five roles — Product Manager, Architect, Project Manager, Engineer, QA Engineer — are one shared GPT-4 backbone differentiated by role profiles (name, profile, goal, constraints) and role-specific tools ([Hong2024MetaGPTMP](#); [sec. 3.1, p. 4](#); GPT-4 backbone at [sec. 4.1, p. 7](#)).
- [MedAgents](#)'s medical experts — question-domain experts, option-domain experts, a report summarizer, voters, and a decision-maker — are all one frozen GPT-3.5 or GPT-4 backbone, differentiated only by system roles and guideline prompts ([Tang2024MedAgentsLL](#); [tab. 12, p. 620](#)).
- [Nascimento2023SelfAdaptiveLL](#)'s five marketplace agents run on one shared frozen GPT-4; despite receiving the same backbone, a seller or buyer role prompt is the only thing splitting them into three sellers and two buyers ([fig. 3, p. 107](#)).

❗ Contradictions

- None yet

Disconfirming Evidence

- Whether role prompts induce genuinely diverse reasoning or only stylistically varied surface text is an open empirical question. The role-prompted setup in [Liu2024ADL](#) does not directly test whether the role-induced distributions differ in reasoning content or only in surface form.
- Role separation does not erase model-level correlation. In [Multi-Agent Debate](#), where a judge and two debaters are all role-prompted from one backbone, the judge favours the debater that shares its backbone model — evidence that the underlying weights still couple agents that the prompts cast as distinct ([Liang2024EncouragingDT](#); [sec. 1, p. 17890](#)).

Open Questions

- What measurement separates content diversity from stylistic diversity in role-prompted outputs? Without one, it is hard to know what role prompting is actually buying.
- Does role prompting scale to harder tasks where the role-induced gloss is thinner than the underlying reasoning the task demands?

Connections

- Underlies [peer rating](#) in [DyLAN](#) — the rater is itself a role-prompted instance and may apply role-influenced rating priors.
- Specialized by [inception prompting](#) in [ChatDev](#) — a matched instructor/assistant pair built from symmetric prompts rather than independent personas.
- The mechanism that instantiates a [role-based strategy](#): dividing work by role is the strategy, and a system prompt on a shared backbone is how each role is realized ([Tran2025MultiAgentCM](#); [sec. 4.3.2, p. 14](#)).
- A multi-agent specialization of the handcrafting method in [agent profiling](#) — assigning a role through a system prompt is how single-agent profiling scales to distinguishing agents on a shared backbone.

Sources

- [Liu2024ADL](#)
- [Chen2024AgentVerseFM](#)

- [Qian2024ChatDevCA](#)
- [Liang2024EncouragingDT](#)
- [Tran2025MultiAgentCM](#)
- [Wang2023AvalonsGO](#)
- [Xu2023ExploringLL](#)
- [Hong2024MetaGPTMP](#)
- [Tang2024MedAgentsLL](#)
- [Nascimento2023SelfAdaptiveLL](#)